

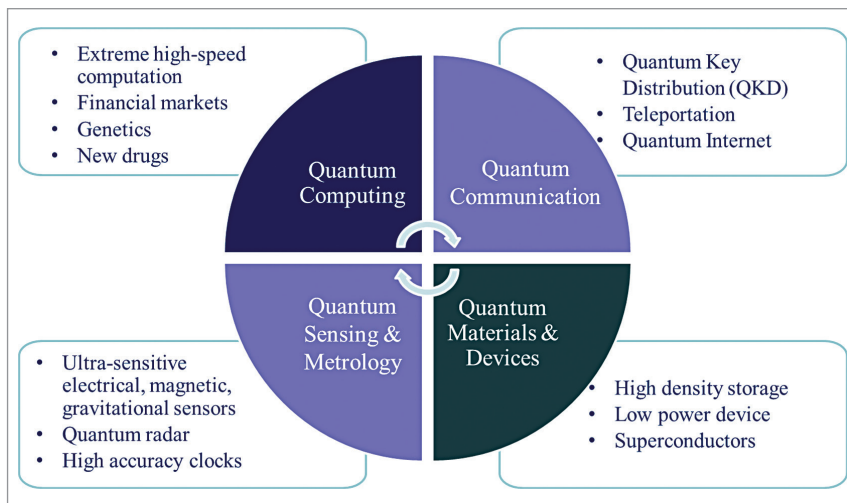


Fig. 1: Verticals and application areas of quantum technologies

market will reach \$106 billion by 2040. Quantum science is set to unlock new frontiers—redefining innovation and placing itself at the core of the next technological revolution.

First quantum revolution

The first quantum revolution, which began in the early 20th century, laid the foundation for modern physics by introducing the principles of quantum mechanics. This era was marked by groundbreaking discoveries such as Max Planck's quantum hypothesis, Einstein's explanation of the photoelectric effect, and Bohr's atomic model, all of which challenged classical physics. The development of wave-particle duality, Heisenberg's uncertainty principle, and quantum electrodynamics further expanded the understanding of the microscopic world. These theoretical advancements led to transformative technologies, including transistors, lasers, atomic clocks, and semiconductors, which became the backbone of modern electronics, computing, and communication.

India has a distinguished legacy in quantum science and technology, with foundational contributions that have shaped global advancements

in the field. Since the early 20th century, the nation has been at the cutting edge of both theoretical and experimental research in quantum mechanics. In the 1920s, the esteemed physicist Satyendra Nath Bose made groundbreaking contributions, establishing the basis for Bose-Einstein statistics and the Bose-Einstein condensate theory. Additionally, C.V. Raman, who was awarded the Nobel Prize in 1930 for his pioneering work on light scattering, also laid down fundamental principles in the field. These contributions have significantly advanced the global understanding of quantum mechanics.

Quantum 2.0

The Second Quantum Revolution, unfolding in the 21st century, harnesses principles such as superposition, entanglement, and coherence to drive innovation. While the first revolution focused on understanding quantum mechanics, this new phase applies those phenomena to four major areas: quantum computing, quantum communication, quantum sensing and metrology, and quantum materials and devices.

Quantum computing is poised to solve complex problems

exponentially faster than classical computers, while quantum cryptography offers unparalleled security for data transmission. Advancements in quantum sensing and metrology are enabling ultra-precise measurements, crucial for fields such as healthcare, defence, and navigation. Progress in quantum materials and devices will drive innovation across the other three verticals. Additionally, quantum communication networks are emerging as the future of secure information exchange. These breakthroughs are shaping the future of science, industry, and national security, making quantum technology a key area of global research and investment.

Core principles of quantum mechanics

Quantum mechanics governs the behaviour of particles at the smallest scales, where classical physics no longer applies. Understanding these principles is key to unlocking the potential of the quantum world.

Superposition. In classical computing, a bit can be either 0 or 1, but a quantum bit (qubit) can exist in both states simultaneously, until measured. As depicted in Fig. 2, the qubit $|\psi\rangle$ is in a superposition of states $|0\rangle$ and $|1\rangle$, and $|\alpha|^2$ and $|\beta|^2$ denote the probability of $|\psi\rangle$ being $|0\rangle$ and $|1\rangle$, respectively, when measured. Imagine flipping a coin—it is neither heads nor tails until it lands. This property, called superposition, allows quantum computers to perform multiple calculations at once, enabling exponential speed-up in problem-solving. However, any measurement collapses the superposition, forcing the qubit into a definite state.

Entanglement. Quantum entanglement is a counterintuitive phenomenon where two particles become so deeply linked that their